How does a network remember the past?

$$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T \rightarrow \mathbf{h}_t \rightarrow \mathbf{c}_t \rightarrow \mathbf{e}_w \rightarrow \alpha_{\tau t} \rightarrow \hat{y}$$

Roadmap

1. Sequences as data — windows, splits and the baseline to beat
2. Recurrence and its gradients — one cell, many steps
3. Memory that lasts — LSTM and GRU
4. Text as a sequence — tokens, embeddings, order
5. Sequence to sequence — and the first form of attention

Practice A after Part 3; Practice B after Part 5.

From fixed inputs to sequences

Sessions 03 and 04: the **architecture** and the **objective** shape the representation $\mathbf{h} = g(\mathbf{x})$.

So far every input had a **fixed size**: 28×28, 32×32, 64×64.

Now the length changes, and the **order** of the steps carries the information.

The question for today

> **How does a network remember the past?**

It sees one step at a time.

- What should it keep from the steps before?
- How does the gradient reach a step that happened long ago?

Part 1 · Sequences as data

1. **Sequences as data** — windows, splits and the baseline to beat
2. Recurrence and its gradients — one cell, many steps
3. Memory that lasts — LSTM and GRU
4. Text as a sequence — tokens, embeddings, order
5. Sequence to sequence — and the first form of attention

> Session 03: the architecture is an assumption about the input. Today the assumption is about **time**.

Order matters

Text, sensors, audio, prices, gait, heart rate.

Shuffle the steps and the meaning can disappear: "not good, but fun" ≠ "good, but not fun".

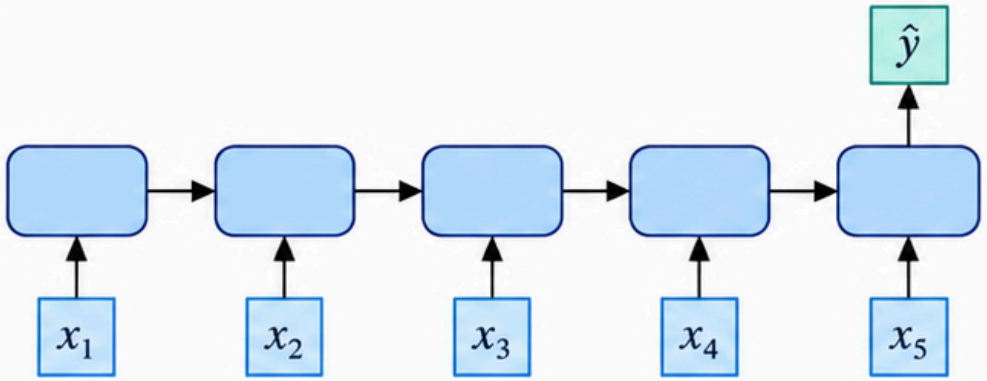
Session 03: shuffling the pixels hurt the CNN, not the MLP. An architecture that ignores order **cannot** use it.

Tasks on sequences

Task	Input → output	Example
many-to-one	sequence → one value	sentiment of a review, next temperature
aligned many-to-many	one output per step	tagging each word, anomaly per reading
sequence-to-sequence	sequence → another sequence	translation, date conversion
one-to-many	one input → sequence	generating text from a start

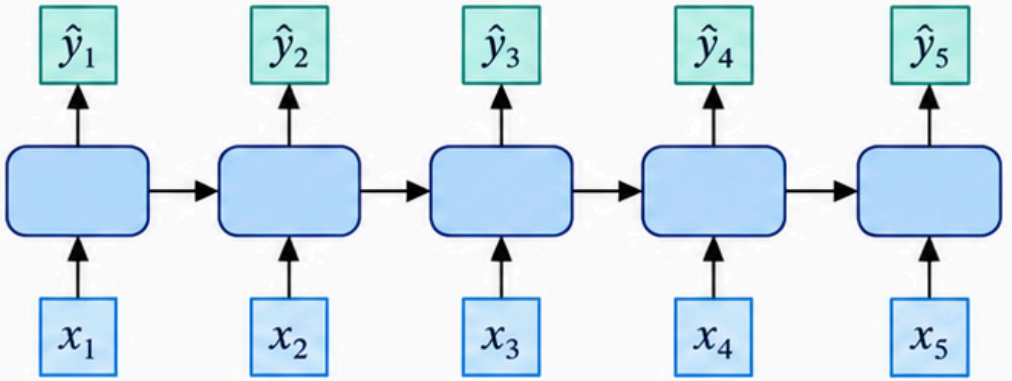
Today: many-to-one in Parts 1–4, sequence-to-sequence in Part 5.

many-to-one



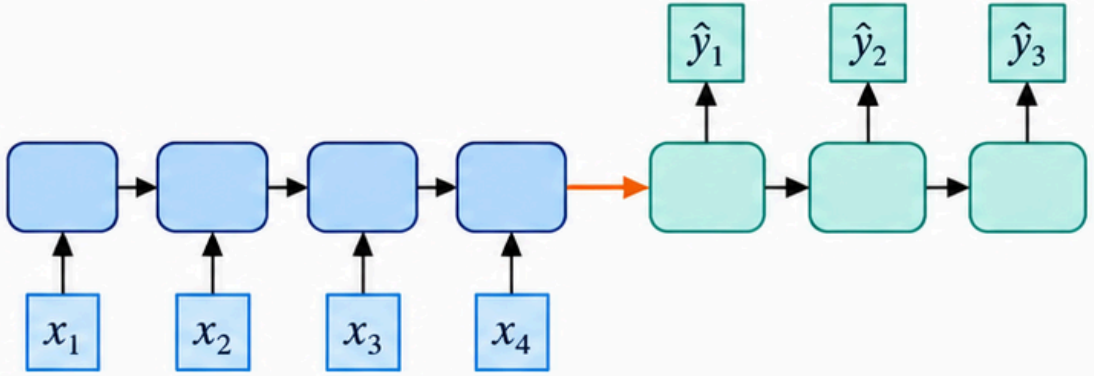
sentiment of a review • temperature in 24 h

aligned many-to-many



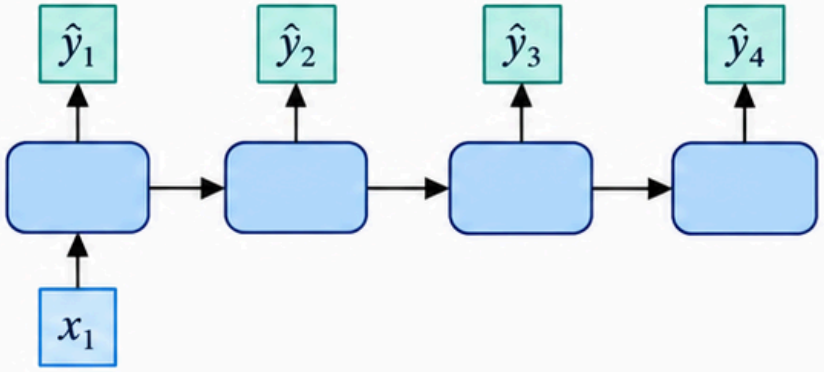
a tag for every word • an anomaly flag for every reading

sequence-to-sequence



translation • date in words → ISO date

one-to-many



generating text from a start

From a series to examples

Jena Climate: 14 variables (temperature, pressure, humidity, wind…) every **10 minutes**, 2009–2016.

A **sliding window** turns one long series into many examples:

$$\underbrace{\mathbf{x}_{t-T+1}, \ldots, \mathbf{x}_t}_{T \text{ past steps}} \longmapsto \; y = \text{temperature at } t + H_{\text{fut}}$$

Notebook 5.1: one reading per hour, $T = 120$ hours (5 days), $H_{\text{fut}} = 24$ hours.

S05-F02 · Windows and the temporal split

The Jena temperature series with three windows and their 24 h targets; train / validation / test as consecutive blocks; no window crosses a cut.

`../figures/s05_f02_windows_split.png` · route: script (notebook 5.1)

Three ways to leak the future

1. **Shuffle before splitting.** Neighboring windows overlap almost completely; one lands in train, its twin in validation.
2. **Normalize with the whole series.** The mean and the scale of the test years enter the training data.
3. **Windows that cross a cut.** The input lives in train, the target in validation.

Each one gives a validation score the test will not repeat.

> The data contract of the S02–S04 integrators, now with time in it: **fit everything on the past only.**

Prediction before experiment

Jena, 5 days of hourly readings → temperature in 24 h.

Commonsense baseline: the temperature in 24 h equals the temperature now.

Flattened MLP: $120 \times 14 = 1680$ inputs, two hidden layers.

Which one has the lower validation MAE? Answer in notebook 5.1.

Why not an MLP or a CNN?

Flattened MLP: parameters grow with T; nothing is shared across time; order is only a position in a vector.

1-D convolution (Session 03 along time): shares weights, but sees a fixed receptive field.

What we want:

- the **same weights at every step**;
- a **summary** of everything seen so far, of fixed size.

Part 2 · Recurrence and its gradients

1. Sequences as data — windows, splits and the baseline to beat
2. **Recurrence and its gradients** — one cell, many steps
3. Memory that lasts — LSTM and GRU
4. Text as a sequence — tokens, embeddings, order
5. Sequence to sequence — and the first form of attention

> Part 1: a series becomes windows, split by time; the baseline sets the bar.

The recurrent cell

$$\mathbf{h}_t = \varphi\big(W_{xh}\,\mathbf{x}_t + W_{hh}\,\mathbf{h}_{t-1} + \mathbf{b}\big), \qquad \mathbf{h}_0 = \mathbf{0}$$

- the **same** W_{xh}, W_{hh}, $\mathbf{b}$ at every step: weight sharing along time;
- $\mathbf{h}_t$ summarizes $\mathbf{x}_{1:t}$ in a vector of fixed size;
- many-to-one: $\hat{y} = \mathbf{w}^\top \mathbf{h}_T + b_{\text{out}}$.

Parameters do not depend on T.

Unrolling through time

S05-A01 · Unrolling and backpropagation through time

The cell rolled up, then unrolled over 8 steps with the same W_{hh} highlighted at every step; the gradient flows back from step 8 and its arrow thins step by step.

`../figures/s05_a01_unrolling_bptt.mp4` · route: Manim (`escenas_s05.py`)

Shapes in PyTorch

```
rnn = nn.RNN(input_size=14, hidden_size=16, batch_first=True)
out, h_T = rnn(x)           # x: (B, T, 14)
# out: (B, T, 16)  every h_t      h_T: (1, B, 16)  the last one
y_hat = head(out[:, -1])   # many-to-one: read h_T
```

Notebook 5.2 writes the cell as a Python loop over t and checks that it gives the same `out` as `nn.RNN`.

Backpropagation through time

The unrolled graph is a **deep network with tied weights**: one layer per step.

$$\frac{\partial \mathbf{h}_T}{\partial \mathbf{h}_t} = \prod_{k=t+1}^{T} \frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_{k-1}} = \prod_{k=t+1}^{T} \operatorname{diag}\big(\varphi'(\mathbf{z}_k)\big)\, W_{hh}$$

Session 02: depth was a product of Jacobians. Here it is the **same matrix**, $T - t$ times.

Vanishing and exploding

$$\left\| \frac{\partial \mathbf{h}_T}{\partial \mathbf{h}_t} \right\| \leq \left(\gamma \, \| W_{hh} \| \right)^{T-t}, \qquad \gamma = \max |\varphi'|$$

- $\gamma \| W_{hh} \| < 1$: the signal from step t **fades geometrically**;
- large $\| W_{hh} \|$: the product can **explode**.

With tanh, $\gamma = 1$ and $\varphi' < 1$ away from zero: fading is the usual case.

Bengio, Simard & Frasconi (1994); Pascanu, Mikolov & Bengio (2013).

S05-F03 · How far back does the gradient reach?

$\|\partial L / \partial \mathbf{x}_t\|$ against t for the trained RNN on Jena, log scale, averaged over a validation batch.

`../figures/s05_f03_rnn_gradient_time.png` · route: script (notebook 5.2)

Prediction before experiment

Same Jena windows, a recurrent network with 16 units.

Does it beat the **commonsense baseline**?
And the flattened MLP?

With a window of **1 day** instead of 5, does it get better, worse, or stay the same?

Answer in notebook 5.2.

Gradient clipping

If $\|\nabla_\theta L\| > c$, rescale it to norm c:

```
loss.backward()
torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
optimizer.step()
```

It stops **explosions**. It does nothing for **fading**: a small gradient stays small.

Session 02: read the gradient norm before choosing a fix.

Truncated BPTT

For very long sequences, backpropagate only k steps and carry the state forward **detached**:

```
h = h.detach()   # the next chunk starts from h, but no gradient flows back into it
```

Cheaper in memory, and it limits **by design** how far back the network can learn.

What we need

A path along time where the gradient is **not** multiplied by W_{hh} at every step.

Session 02 had the answer for depth:

$$\mathbf{x}_{l+1} = \mathbf{x}_l + F(\mathbf{x}_l) \qquad \Rightarrow \qquad \frac{\partial \mathbf{x}_{l+1}}{\partial \mathbf{x}_l} = I + \frac{\partial F}{\partial \mathbf{x}_l}$$

An **additive** path. Part 3 puts one along time.

Part 3 · Memory that lasts: LSTM and GRU

1. Sequences as data — windows, splits and the baseline to beat
2. Recurrence and its gradients — one cell, many steps
3. **Memory that lasts** — LSTM and GRU
4. Text as a sequence — tokens, embeddings, order
5. Sequence to sequence — and the first form of attention

> Part 2: the repeated product of Jacobians wipes out the past.

An additive path for memory

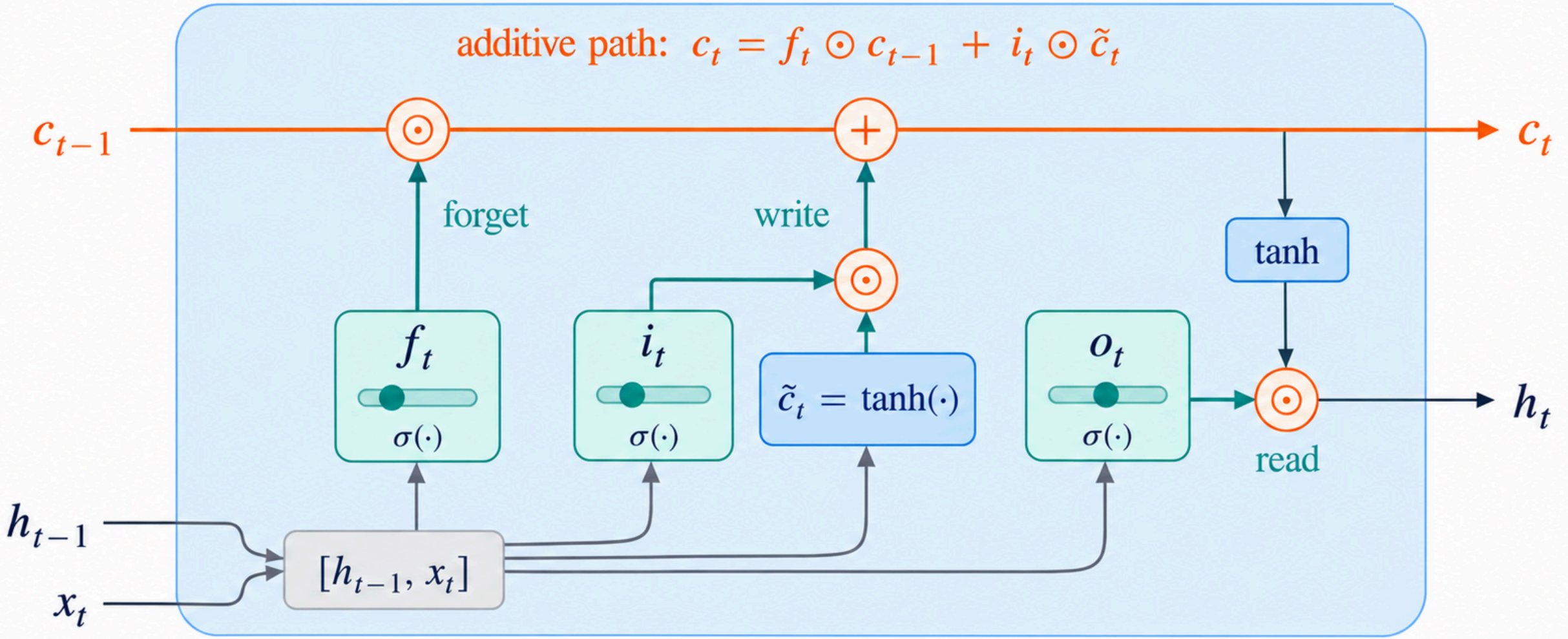
The LSTM keeps a second state, the **cell** $\mathbf{c}_t$:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \qquad \tilde{\mathbf{c}}_t = \tanh\left(W_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c\right)$$

Addition instead of multiplication by W_{hh}: the residual of Session 02, **along time**.

With $\mathbf{f}_t \approx \mathbf{1}$: $\partial \mathbf{c}_t / \partial \mathbf{c}_{t-1} \approx I$.

Hochreiter & Schmidhuber (1997); forget gate: Gers, Schmidhuber & Cummins (2000).



with $f_t \approx 1$: $\dfrac{\partial c_t}{\partial c_{t-1}} \approx I$ — the residual of Session 02, along time

Three gates

$$\mathbf{f}_t, \mathbf{i}_t, \mathbf{o}_t = \sigma\big(W_{f,i,o}\,[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_{f,i,o}\big) \in (0,1)^{d_h}$$

Gate	Decides
forget $\mathbf{f}_t$	how much of $\mathbf{c}_{t-1}$ to keep
input $\mathbf{i}_t$	how much of the new candidate to write
output $\mathbf{o}_t$	how much of the memory to show: $\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$

A gate is a **learned multiplier that depends on the input**. Common trick: forget bias initialized to 1, so memory is kept from the start.

The memory in motion

S05-A02 · Writing, keeping and reading a memory

A value written into $\mathbf{c}_t$ at step 2, kept by $\mathbf{f}_t \approx 1$, read by $\mathbf{o}_t$ at step 7. The same input in a plain RNN fades step after step.

`../figures/s05_a02_lstm_memory.mp4` · route: Manim (`escenas_s05.py`)

GRU

Two gates, no separate cell (Cho et al., 2014):

$$\mathbf{h}_t = (1 - \mathbf{u}_t) \odot \mathbf{h}_{t-1} + \mathbf{u}_t \odot \tilde{\mathbf{h}}_t$$

- update gate $\mathbf{u}_t$: interpolates old and new; the additive path is still there;
- reset gate $\mathbf{r}_t$: how much of $\mathbf{h}_{t-1}$ enters the candidate $\tilde{\mathbf{h}}_t$.

Three blocks of weights instead of four: about **3/4** of the LSTM parameters. Often similar results; choose on validation, not by habit.

Prediction before experiment

Notebook 5.3: LSTM and GRU on the same Jena windows, same hidden size, same budget.

Do they beat the RNN and the baseline? By how much?

Does the gradient-vs-time plot of F03 change **shape**, or only height?

Stacked and bidirectional

Stacked: $\mathbf{h}_t$ of layer 1 is $\mathbf{x}_t$ of layer 2. `nn.LSTM(..., num_layers=2)`.

Bidirectional: one pass forward, one backward, states concatenated. `bidirectional=True`.

Bidirectional reads the **whole input** before predicting:

- text classification: fine, the review is complete;
- forecasting: fine *inside* the window, never beyond its end.

Regularizing recurrent networks

- dropout on the inputs and outputs of the cell;
- dropout on the recurrent connection with the **same mask at every step** (Gal & Ghahramani, 2016); a new mask per step destroys the memory;
- weight decay and early stopping, as in Session 02;
- gradient clipping, always on.

`nn.LSTM(dropout=p)` applies dropout only between stacked layers, not on the recurrent connection.

Practice A — Recurrent networks on Jena

Worked notebooks, one change per cell, GPU recommended:

- 5.1 · Series and windows: the three leaks, shown and fixed; baseline; flattened MLP; 1-D CNN
- 5.2 · RNN and BPTT: the cell as a loop, gradient vs t, clipping, 1 day vs 5 days
- 5.3 · LSTM and GRU: same windows, same plot, stacking, recurrent dropout

> Keep one table: model, parameters, validation MAE, MAE minus baseline.

Part 4 · Text as a sequence

1. Sequences as data — windows, splits and the baseline to beat
2. Recurrence and its gradients — one cell, many steps
3. Memory that lasts — LSTM and GRU
4. **Text as a sequence** — tokens, embeddings, order
5. Sequence to sequence — and the first form of attention

> Part 3: an additive path keeps the signal over long windows. Now the steps are **words**.

From text to integers

1. **Tokenize**: split the text into units.
2. **Vocabulary**: the most frequent tokens of the **training set** only; the rest map to `<unk>`.
3. **Ids**: each token → an integer; special tokens `<pad>`, `<unk>`.

Unit	Vocabulary	Sequence length	Unknown words
characters	tiny	very long	none
words	large	short	many
subwords (BPE)	chosen	in between	none — Session 06

S05-F05 · The text pipeline

A sentence → tokens → ids → a padded batch with its mask → a tensor of embeddings (B, T, d_e).

`../figures/s05_f05_text_pipeline.png` · route: generator

Embeddings

$$\mathbf{e}_w = E[w], \qquad E \in \mathbb{R}^{|V| \times d_e}$$

`nn.Embedding(|V|, d_e)`: one-hot × matrix, done as a lookup (Session 04).

- learned **with the task**, from random;
- or initialized from word2vec or GloVe, trained on other text.

Distance in E comes to mean "similar **for this task**".

Batches of different lengths

Reviews have 10 words or 1 000.

- **pad** to the longest in the batch with `<pad>`, and keep a **mask**;
- the model must ignore the padding: masked mean, or `pack_padded_sequence` for the LSTM;
- **truncate** long texts to a maximum length: a design choice to report;
- group similar lengths in a batch to waste less compute.

A classic silent bug: averaging embeddings including the padding. Short reviews get pulled toward the `<pad>` vector.

Three classifiers, same budget

Model	Sees order?	Reads
bag of embeddings	no	$\frac{1}{T}\sum_t \mathbf{e}_{w_t}$ → linear layer
1-D CNN over embeddings	locally, k words	filters over windows of k words, max over time
LSTM	all of it	$\mathbf{h}_T$ (or a pooling of every $\mathbf{h}_t$)

Same embedding size, same training budget, IMDB (25 000 reviews for training, 25 000 for test).

S05-F06 · What each text model sees

The same sentence through the three models: an average with no order, sliding windows of 3 words, a chain of states.

`../figures/s05_f06_three_text_models.png` · route: generator

What the embedding learned

After training on sentiment, look at the nearest neighbors of a few words in E (cosine similarity, as in Session 04).

The neighbors are grouped by **polarity**, not by meaning: an adjective of praise sits next to other praise, whatever it describes.

An embedding is useful **for its task** (Session 04).

S05-F07 · Nearest neighbors in the learned embedding

Six query words and their five nearest neighbors in E after training on IMDB.

`../figures/s05_f07_embedding_neighbors.png` · route: script (notebook 5.4)

Prediction before experiment

IMDB. Shuffle the words of every **test** review; the models were trained on normal text.

Bag of embeddings: what happens to its accuracy?

(Think about the formula before answering.)

Answer in notebook 5.4.

LSTM: how much does it lose?

What does the gap say about whether **order matters for sentiment**?

When order matters

Sentiment is close to a bag of words: which words appear says most of it.

Order matters for negation ("not good"), contrast ("good acting, but…"), irony, and anything with structure: dates, code, translation.

The **task** decides whether the order-aware model is worth its cost.

The limit of a single vector

Many-to-one keeps only $\mathbf{h}_T$, or a pooling of all the $\mathbf{h}_t$.

For a sentiment score, one vector is enough.

For a **translation**, that one vector must carry the whole input sentence to a model that writes another one.

Part 5 · Sequence to sequence and attention

1. Sequences as data — windows, splits and the baseline to beat
2. Recurrence and its gradients — one cell, many steps
3. Memory that lasts — LSTM and GRU
4. Text as a sequence — tokens, embeddings, order
5. **Sequence to sequence** — and the first form of attention

> Part 4: one vector summarizes the sequence. Enough to classify, not to rewrite.

Encoder and decoder

Encoder: an RNN reads $\mathbf{x}_1, \dots, \mathbf{x}_T$ and ends in $\mathbf{h}_T$.

Decoder: another RNN starts from $\mathbf{s}_0 = \mathbf{h}_T$ and writes $y_1, \dots, y_{T'}$, one token at a time:

$$\mathbf{s}_\tau = \mathrm{RNN}(\mathbf{s}_{\tau-1}, \mathbf{e}_{y_{\tau-1}}), \qquad p(y_\tau \mid y_{<\tau}, \mathbf{x}) = \mathrm{softmax}(W_o \mathbf{s}_\tau)$$

The encoder–decoder of Session 04, now in time. Sutskever, Vinyals & Le (2014); Cho et al. (2014).

S05-F08 · Sequence to sequence

Encoder RNN over "diecinueve de octubre de 2026", the single vector $\mathbf{h}_T$ highlighted, decoder RNN writing `2026-10-19` character by character.

`../figures/s05_f08_seq2seq.png` · route: generator

Teacher forcing

Training: the decoder receives the **true** previous token $y_{\tau-1}$. All steps are trained in one pass; the loss is the cross-entropy of Session 01, summed over τ.

Inference: it receives **its own** previous prediction. Errors accumulate: exposure bias.

Decoding: greedy (take the most likely token); beam search keeps the k best partial outputs.

The bottleneck

Everything the decoder knows about the input passes through **one vector** $\mathbf{h}_T$, of fixed size, whatever the input length.

Translation quality of the plain encoder–decoder **falls with the length** of the sentence (Cho et al., 2014).

Part 2 again: the beginning of the input is many steps away from the end.

Look back instead of remembering

At each output step τ, the decoder **looks at every encoder state**:

$$\alpha_{\tau t} = \mathrm{softmax}_t\left(\mathrm{score}(\mathbf{s}_{\tau-1}, \mathbf{h}_t)\right), \qquad \mathbf{a}_\tau = \sum_{t=1}^{T} \alpha_{\tau t}\, \mathbf{h}_t$$

The context $\mathbf{a}_\tau$ changes at every output step and enters the decoder with $\mathbf{e}_{y_{\tau-1}}$.

Bahdanau, Cho & Bengio (2015).

Attention in motion

S05-A03 · Decoding a date with attention

At each output character: scores over the input words, softmax into $\alpha_{\tau t}$, the weighted sum $\mathbf{a}_\tau$, the emitted character. The matrix of weights fills row by row.

`../figures/s05_a03_seq2seq_attention.mp4` · route: Manim (`escenas_s05.py`)

Score functions

Score	$\text{score}(\mathbf{s}, \mathbf{h})$	Source
additive	$\mathbf{v}^\top \tanh(W_s \mathbf{s} + W_h \mathbf{h})$	Bahdanau et al. (2015)
dot product	$\mathbf{s}^\top \mathbf{h}$	Luong et al. (2015)
scaled dot product	$\mathbf{s}^\top \mathbf{h} / \sqrt{d}$	Vaswani et al. (2017)

The scaled dot product is the one that survives in **Session 06**.

S05-F09 · Attention as an alignment

The weights $\alpha_{\tau t}$ for three dates: output characters (rows) × input words (columns). The year digits look at the year, the month digits at the month name.

`../figures/s05_f09_attention_alignment.png` · route: script (notebook 5.5)

Prediction before experiment

Spanish dates in words → ISO format. Seq2seq with and without attention, same size, same training.

Test on the **longest** written forms ("veintinueve de septiembre de mil novecientos noventa y ocho"). Which model degrades?

When the decoder writes the two **month** digits, where will its attention be?

Answer in notebook 5.5.

What attention changed

- the decoder reaches **any input step directly**: path length 1 instead of T;
- the bottleneck vector disappears;
- the weights read as an **alignment** — with the care of Session 03: a map is a clue, not an explanation.

Keep the attention, drop the recurrence?

The encoder is still recurrent: step t waits for step $t-1$, so training cannot run the steps in parallel.

If every step can look at every other step directly…

> …is the recurrence still needed? **Session 06.**

Practice B — Text and sequence to sequence

Worked notebooks:

- 5.4 · Text and embeddings: vocabulary from train, padding and masks, bag of embeddings vs 1-D CNN vs LSTM, shuffled words, neighbors in E · GPU
- 5.5 · Seq2seq and attention: dates in words → ISO, with and without attention, accuracy vs length, the attention map · CPU

> Write down for each text model: parameters, test accuracy, accuracy with shuffled words.

Integrating exercise — Bike Sharing

5.6 · Integrating exercise · hourly bike rentals, weather and calendar

- temporal split and windows **without leakage**;
- your recurrent model (LSTM or GRU), logged to **MLflow**;
- MAE at 1, 6, 12 and 24 hours ahead, against the baseline;
- `forecast_raw`: a forecast from raw hourly records that rejects those that break the contract.

> How many hours ahead does your network still beat the obvious answer?

Choosing a sequence model

Model	Sees	Parameters grow with T?	Enough when
baseline	the last value (or season)	—	always the first row of the table
MLP on the window	T steps, no order	yes	short windows, few variables
1-D CNN	a receptive field	no	local patterns
RNN	everything, fading	no	short dependencies
LSTM / GRU	everything, kept	no	long dependencies
seq2seq + attention	every input step, at every output step	no	output is another sequence

What each choice assumes

	Assumes	Pays with
the window	the past beyond T does not matter	what happened before T
the RNN	the past fits in $\mathbf{h}_t$	the fading gradient
the LSTM	gates can learn what to keep	4× the RNN parameters, sequential training
bag of embeddings	order does not matter	negation, structure
attention	any step may matter for any output	a weight for every pair of steps

S05-F10 · How each model reaches the past

Window, RNN, LSTM and attention side by side: how much of step 1 is still available when the model makes its prediction at step T.

`../figures/s05_f10_summary.png` · route: generator

Exit question

A hospital has two years of hourly bed occupancy and wants a **48-hour** forecast.

Choose two and justify:

- a commonsense baseline with the weekly pattern;
- an LSTM with a 48-hour window;
- an LSTM with a 2-week window;
- a bidirectional LSTM over the window;
- first check that the split keeps the last months for test.